Advanced Spectroscopic Techniques

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Advanced Spectroscopic Techniques (A11564)

Short Title:	Adv. Spectroscopic Techniques		
Faculty	Science and Computing		
Department	Science		
Cluster	Analytical Science		
Author	KGRENNAN		
Credits	10	**Level:**	Postgraduate

Description of Module / Aims

This module will provide the student with an advanced understanding of the theory, practices and instrumentation associated with advanced spectroscopic techniques. The techniques covered include molecular (UV-Vis, IR, Raman, NMR, MS) and atomic (AAS, AES, AFS) spectroscopies, focusing on state-of-the-art developments in their practices and instrumentation, as well as issues to deal with the spectroscopic identification of samples in complex matrices. The learner will be exposed to the application of advanced separation techniques in a wide range of environments focusing mainly, but not exclusively, on applications in the pharmaceutical and biopharmaceutical industries.

Programmes

`SCIE-0051`	Certificate in Advanced Spectroscopic Techniques (`WD_SSPEC_MA`)	5 / 0 / M
`SCIE-0051`	MSc in Analytical Science with Quality Management (`WD_SANSC_R`)	1 / 0 / E
`SCIE-0051`	PG Dip in Analytical Science with Quality Management (`WD_SANSC_G`)	1 / 0 / E

Indicative Content

- Advanced options and state-of-the-art developments in infra-red (IR) & UV-Vis instrumentation, e.g. diode array
- Evaluation of IR sampling techniques, e.g. multi- & single-attenuated total reflectance (ATR), diffuse reflectance, transmission systems (liquid & solid)
- The use of vibrational (IR & Raman) spectroscopy for the analysis of polymorphs, hydrates and solvates (cf. their quantification and identification) and for in situ characterisation, including Process Analytical Technology (PAT)
- Mass-Spectrometry interpretation (GC & LC), fragmentation patterns, mass analysers, Maldi methodologies
- Advanced options & state-of-the-art developments in atomic absorption (AAS), atomic emission (AES) and atomic fluorescence (AFS) spectroscopies, including quantitative analysis for metals in complex matrices & interference correction methods
- Inductively coupled plasma (ICP) techniques, e.g. ICP-MS and ICP-AES
- Review of advanced interpretation of NMR methods for structural elucidation, e.g. proton, carbon & 2-D techniques
- Advanced options and state-of-the-art developments in Nuclear Magnetic Resonance (NMR) spectroscopy, e.g. on-line separations, quantitation, solid-state NMR
- Principles and practices of polarimetry, fluorescence & phosphorescence techniques
- Selected applications of all of the above in the pharmaceutical and biopharmaceutical industries, incorporated into each area above
- Laboratory practicals demonstrating ability to operate the above techniques

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the principles, practices and instrumentation associated with advanced spectroscopic techniques.
2. Solve sample preparation issues relating to the use of advanced spectroscopic techniques.
3. Solve method development issues concerned with mass spectrometry and related hyphenated techniques.
4. Set up and calibrate associated laboratory instrumentation for pharmaceutical and biopharmaceutical sample analyses.
5. Select appropriate advanced data analysing, synthesising and summarising skills in spectroscopic techniques.

Learning and Teaching Methods

- Lectures.
- Practicals.
- Demonstrations.
- E-learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Practical	50%	2,3,4
In-Class Assessment	50%	1,5

Assessment Criteria

<40%: The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Will display a lack of competence with regards to practical and resulting data analysis skills.

40%-59%: The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regards to associated practical and resulting data analysis skills.

60%-69%: As well as the above, the learner will demonstrate reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to associated practical and resulting data analysis skills.

70%-100%: All of the above to excellent levels. In addition, the learner will demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to associated practical and resulting data analysis skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Practical		24
Independent Learning		222

Essential Material

- "Advanced Spectroscopic Techniques module Moodle page." http://moodle.wit.ie

Requested Resources

- Room Type: Chemistry Lab
- Room Type: Computer Lab